

PRESENTATION ON

LifeOS

Smart Life Decision Operating System

Final Year Project



PRESENTED BY

Badal Kumar

Roll No: 10 · Session: 2023–26

Coordinator & HOD: **Dr. Dinesh Manjhi**

Guided by: **Ajit Sir**

Department of Computer Application

Patna College, Patna

THE PROBLEM

"We make **35,000 decisions** a day — but how many do we **learn from**?"

✗ No Structure

Decisions made without documentation or context

✗ No Feedback

No system to track what actually happened

✗ No Learning

Same mistakes repeated because patterns are invisible

THE SOLUTION

An operating system for your **life decisions**



Log

Decisions with context, alternatives & expected outcomes



Track

Schedule check-ins to record actual results over time



Advise

AI advisor that knows your personal decision history



Discover

Auto-detect behavioral patterns and blind spots

"Decide better. Learn faster. Live intentionally."

CORE FEATURES

What LifeOS Does

 **Decision Journal**

Structured logging with categories, alternatives, confidence levels, and expected metrics

 **Outcome Tracking**

Scheduled check-ins to measure satisfaction, record reflections, and track lessons learned

 **AI Advisor**

Conversational AI with SSE streaming that knows your full decision history

 **Pattern Detection**

Auto-discovers timing effects, overconfidence bias, and category-specific tendencies

 **Analytics**

Visualize decision quality, satisfaction trends, and category breakdowns over time

 **Memory System**

AI extracts key facts from your reflections for personalized long-term guidance

TARGET USERS

Who Is This For?



Students

Career paths, internships,
course selection



Professionals

Job offers, salary, project
prioritization



Entrepreneurs

Strategy, investments,
partnerships

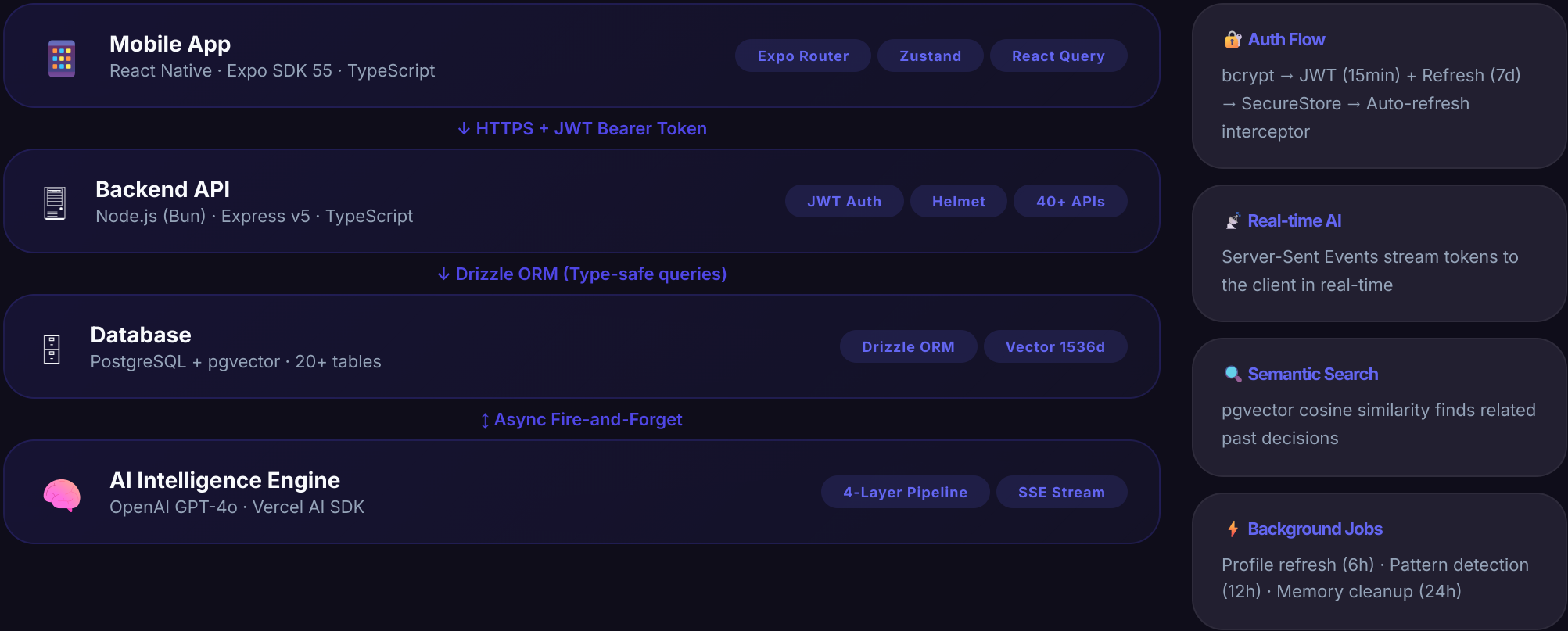


Everyone

Health, purchases,
relationships, lifestyle

ARCHITECTURE

System Architecture & Technology



Database Design

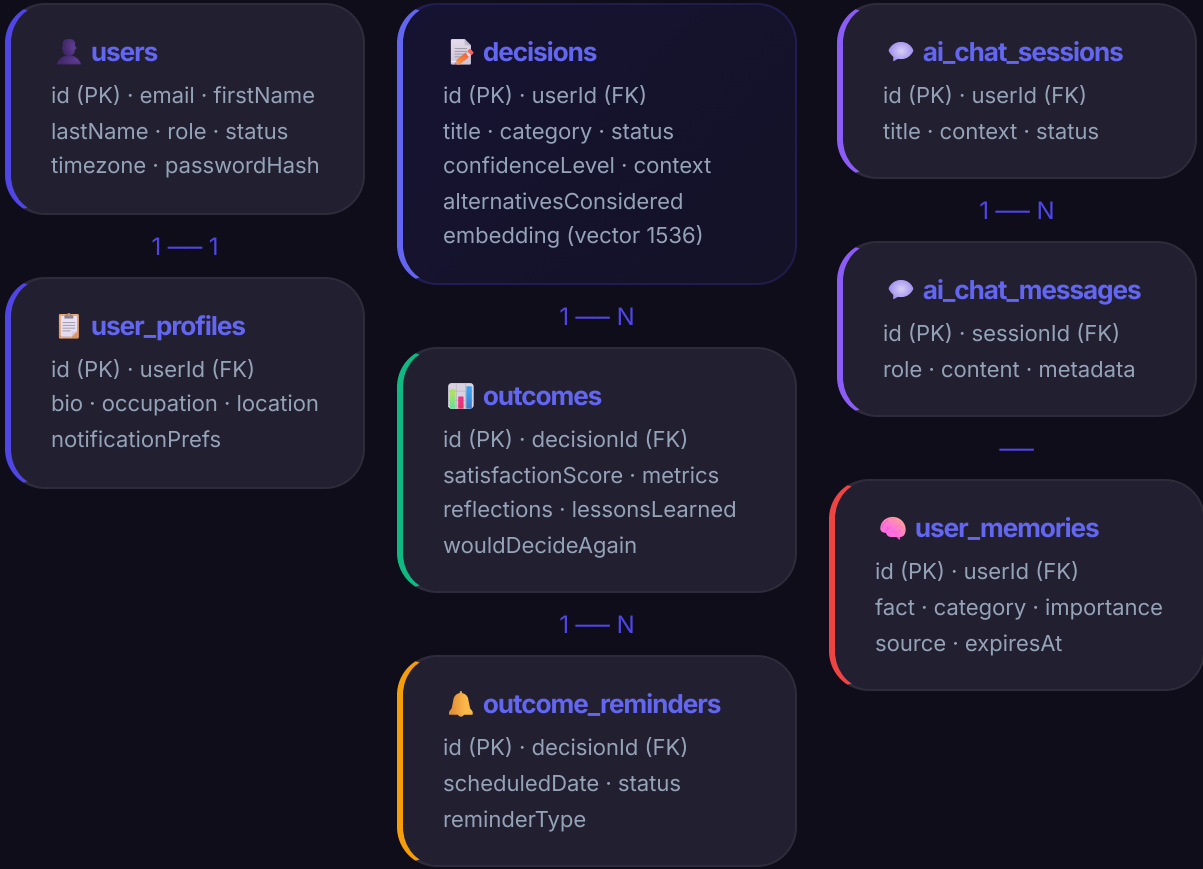
20+ tables across 10 schema modules — all type-safe via Drizzle ORM

Module	Key Tables	Purpose
Users	users, user_profiles	Auth, preferences, settings
Auth	refresh_tokens	JWT token management
Decisions	decisions, frameworks, templates	Core decision data + attachments
Outcomes	outcomes, outcome_reminders	Result tracking over time
AI	ai_chat_sessions, ai_chat_messages	Conversational AI history
Intelligence	user_decision_profiles, decision_patterns, user_memories	AI-computed behavioral data
Analytics	analytics_snapshots, user_insights	Periodic metrics + insights
Notifications	notifications	User alerts & reminders

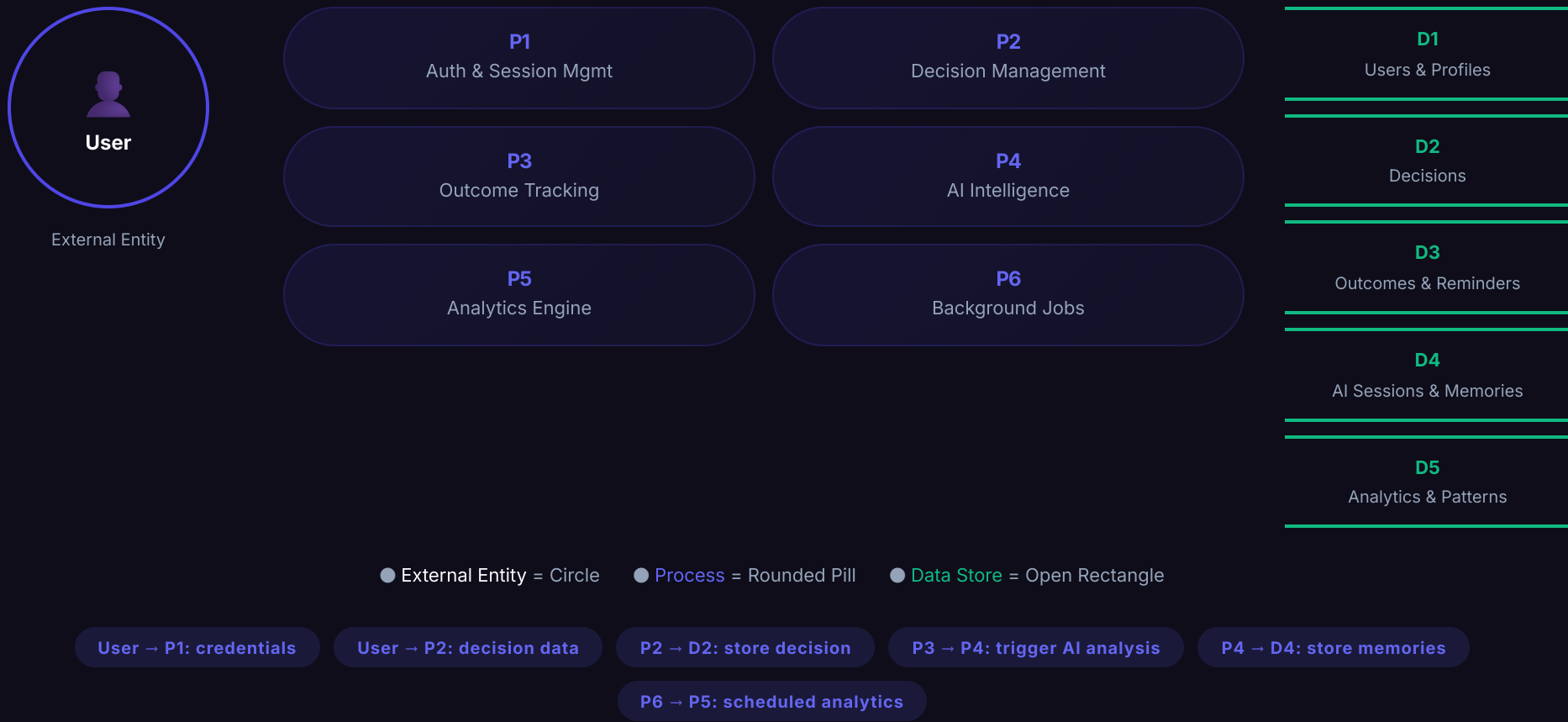
pgvector

decisions.embedding — 1536-dim vectors for semantic similarity search

Entity Relationship Diagram

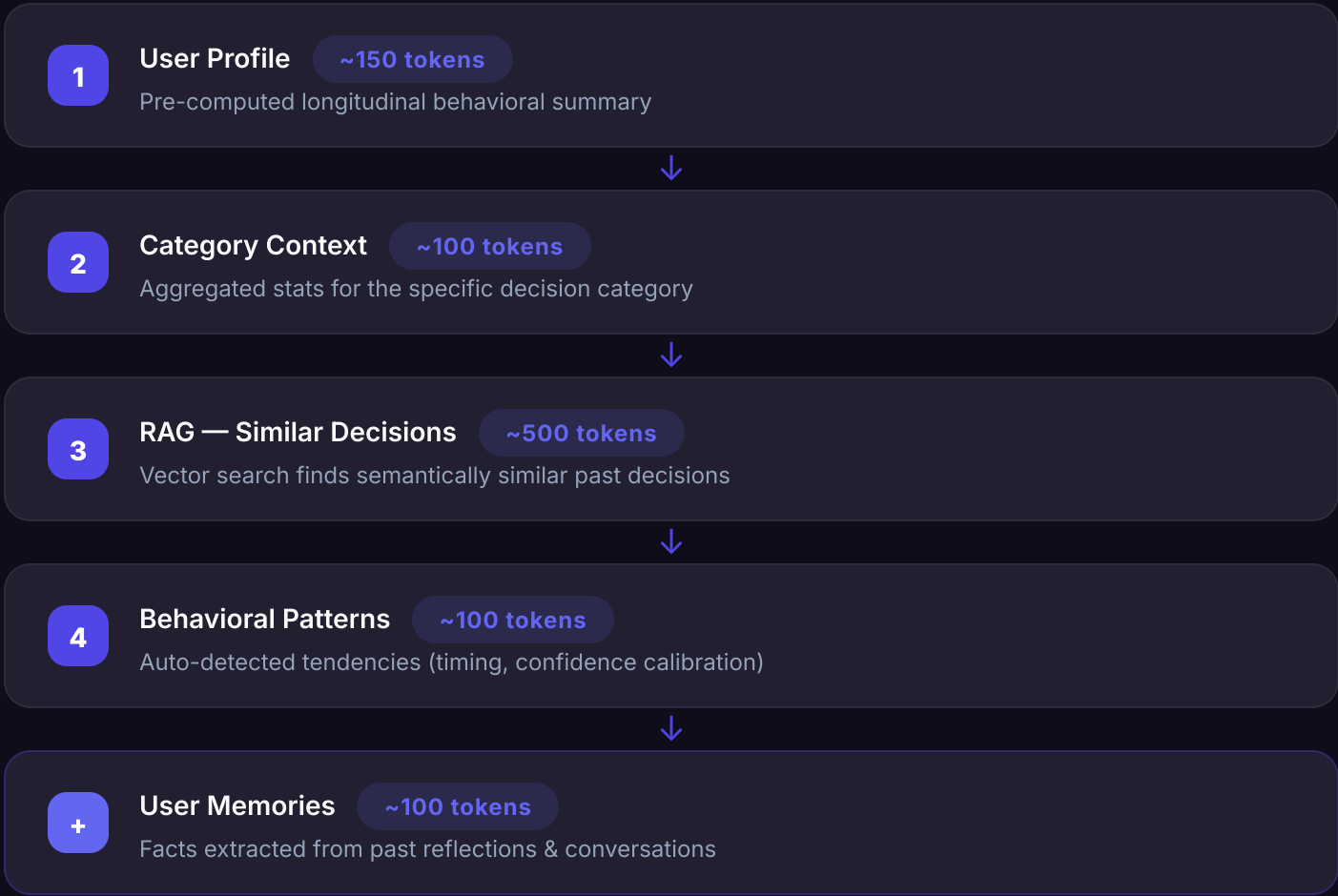


Data Flow Diagram (Level 1)



AI Context Pipeline

Every AI response is personalized through a 4-layer context assembly



API Design

MODULE	ENDPOINTS
Auth	register, login, refresh, logout
Users	getMe, updateMe, profile CRUD, delete
Decisions	list, get, create, update, delete
Outcomes	list, create, update, checkins (3)
Analytics	summary, patterns, quality, insights
AI	chat (SSE), sessions, analyze, patterns
Frameworks	list, get, create, delete
Templates	list, get, create, delete

Key Design Decisions

- **JWT Auth** — Bearer tokens on all routes
- **Pagination** — All list endpoints
- **SSE Streaming** — Real-time AI chat
- **Fire & Forget** — Async AI triggers
- **Soft Deletes** — Never lose data

DESIGN

Design System

Minimalist, calm, premium — designed to reduce decision anxiety

Color Palette

- Background
#F9FAFB
- Accent (Indigo)
#4F46E5
- Text Primary
#111827
- Success
#10B981
- Warning
#F59E0B
- Danger
#EF4444

UI Principles

Typography

Inter font · Bold headings (700-800) · Regular body (400)

Components

Pill buttons (9999px radius) · Rounded cards (20px) · No heavy borders

Interactions

Pull-to-refresh · Skeleton loaders · Gradient CTAs · Smooth transitions

SHOWCASE

App Screens



Auth Flow

Login & Register with gradient CTAs,
form validation, secure token storage



Dashboard

Greeting, action cards, quick actions,
AI reflection, activity feed



Decision List

Search, category filters, status pills,
confidence rings, FAB



Decision Detail

Segmented tabs, outcome timeline, AI
analysis button



Decision Wizard

3-step form with template autofill &
confidence slider



AI Advisor

Chat interface with streaming,
markdown, session history

DEEP DIVE

Decision Wizard

Step 1 — Foundations

- Template autofill from backend
- Title input
- Category selection grid (10 categories)
- Context/description textarea

Step 2 — Context

- Background & reasoning textarea
- Numbered alternative inputs
- Add/remove alternatives
- Pros & cons for each option

Step 3 — Calibration

- Expected outcome metrics
- Target date picker
- Confidence slider (1-10)
- Submit with validation

- useReducer
- React Query Mutations
- Form Validation
- Optimistic UI

Authentication & Security

LAYER	IMPLEMENTATION
Passwords	bcrypt hashing with salt
Access Tokens	JWT — 15 min expiry
Refresh Tokens	Secure random — 7 day expiry
Mobile Storage	expo-secure-store (encrypted keychain)
API Protection	Bearer token middleware on all routes
Auto-Refresh	Axios interceptor: 401 → refresh → retry
Data Safety	Soft deletes — never hard-delete user data

Auth Flow

```
User → Login (email/pass)
→ Server validates → bcrypt compare
→ Issue accessToken + refreshToken
→ Store in SecureStore
→ Attach to every API request
→ 401? Auto-refresh → retry
```

AUTOMATION

Background Intelligence

6h

PROFILE REFRESH

Recompute user behavioral
profile from all decisions &
outcomes

12h

PATTERN DETECTION

Discover timing effects,
overconfidence, and category
biases

24h

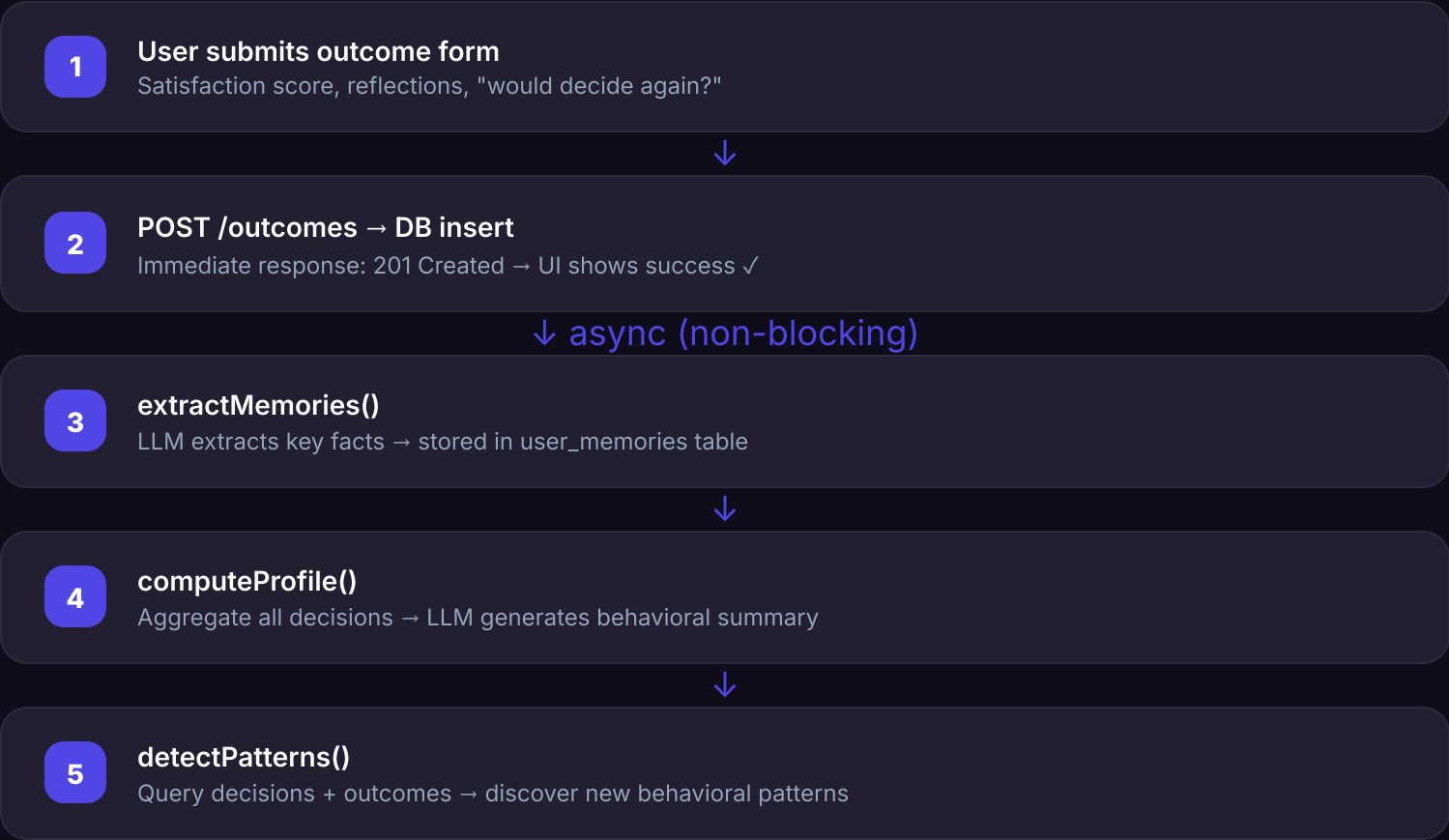
MEMORY CLEANUP

Remove expired memories, cap
at 50 highest-importance per
user

Also triggered on-demand when recording outcomes — **fire-and-forget async**

FLOW

Data Flow: Record an Outcome



SCALE

Project Metrics

20+

DB TABLES

40+

API ENDPOINTS

10+

APP SCREENS

20+

COMPONENTS

5

AI SERVICES

11K+

LINES OF CODE

15+

TECHNOLOGIES

10

SCHEMA
MODULES

LESSONS

Challenges & Solutions

CHALLENGE	SOLUTION
AI chat response latency	SSE streaming for real-time token delivery
Data consistency for AI analysis	Fire-and-forget async triggers (non-blocking)
JWT token expiry during app use	Axios interceptor auto-refresh + retry
Complex decision form state	useReducer with typed action dispatch
IoT hardware integration for health tracking	Future R&D — custom wearable devices with BLE protocol

Future Scope



IoT Health Tracking

THE BIG VISION

Track life activities through **smart wearable sensors** — heart rate, SpO2, sleep quality, walking & running metrics.

 LifeOS Smart Device

Launch our own tracking hardware — a wristband that monitors **heart, SpO2, sleep, steps**. All data feeds into LifeOS to generate an overall **Life Habits Score** — AI correlates your health data with decision quality.

-  Heart Rate

 SpO2

 Sleep

 Activity

 Life Score



Dark Mode

Full midnight blue theme



Push Notifications

Check-in reminders via Expo Push



Decision Sharing

Share anonymized decisions socially



Advanced Analytics

Charts, heatmaps, quality timeline



Voice Input

Speech-to-text for decisions



Export / Backup

CSV/PDF decision journal export

CONCLUSION

Decide better.
Learn faster.
Live intentionally.

LifeOS closes the gap between making decisions and learning from them.
It's the first platform that builds a personalized AI model of *your* decision-making patterns.

[Live Demo →](#)

Thank You!

[Questions?](#)